



14A CHEMICAL KINETICS

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 14

1. Express reactions rates in relation to reaction stoichiometry.
2. Use experimental data to write rate laws for chemical reactions.
3. Predict the concentration and time values of reactions using first- and second-order integrated rate laws.
4. Graph data to identify the order of a reaction.
5. Define and apply the terms half-life, first-order reaction, and second-order reaction.
6. Explain collision theory using appropriate terminology.
7. Apply the Arrhenius equation to calculate activation energy and/or the value of the rate constant at various temperatures.
8. Define and discuss reaction mechanisms.
9. Write rate laws for elementary steps.
10. Identify intermediates in a reaction mechanism.
11. Discuss how catalysts work to speed up chemical reactions

Rates of Reactions

14.1

The rate of a chemical reaction is often measured as the **change in the concentration** of a reactant or product **per unit time**.

- Common units: $\frac{\text{M}}{\text{s}}$ or $\frac{\text{mol}}{\text{L}\cdot\text{s}}$ or ...

$$\text{Rate} = \frac{\Delta[\text{concentration}]}{\Delta\text{time}}$$

Factors that affect reaction rates

- Particle size of solid reactants
- Concentration of reactants*
- Temperature*
- Nature of the reactants
- Presence of a catalyst

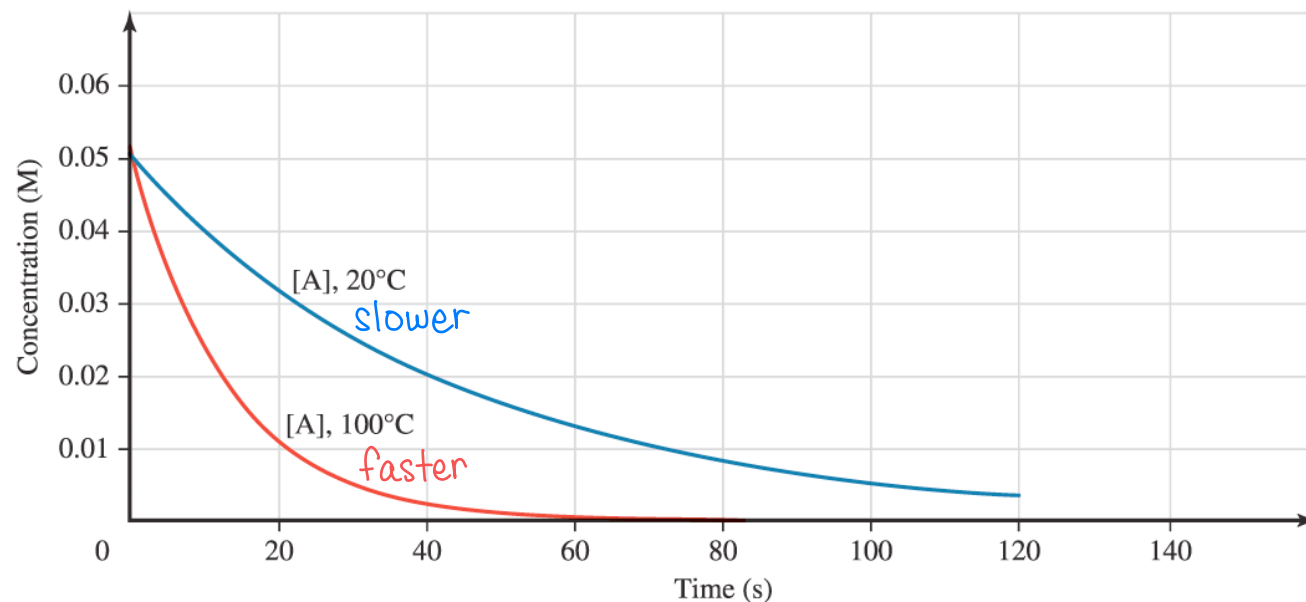
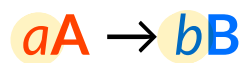


FIGURE 14.1

Rates of Reactions

14.1

The rate of the generic reaction below can be expressed as either:



stoichiometric coefficients

- the rate at which $[A]$ decreases or
- the rate at which $[B]$ increases.

$$\text{Rate} = \frac{\Delta[\text{concentration}]}{\Delta \text{time}}$$

$$\text{Rate} = - \underbrace{\frac{1}{a} \cdot \frac{\Delta[A]}{\Delta t}}_{\text{negative}} = + \underbrace{\frac{1}{b} \cdot \frac{\Delta[B]}{\Delta t}}_{\text{positive}}$$

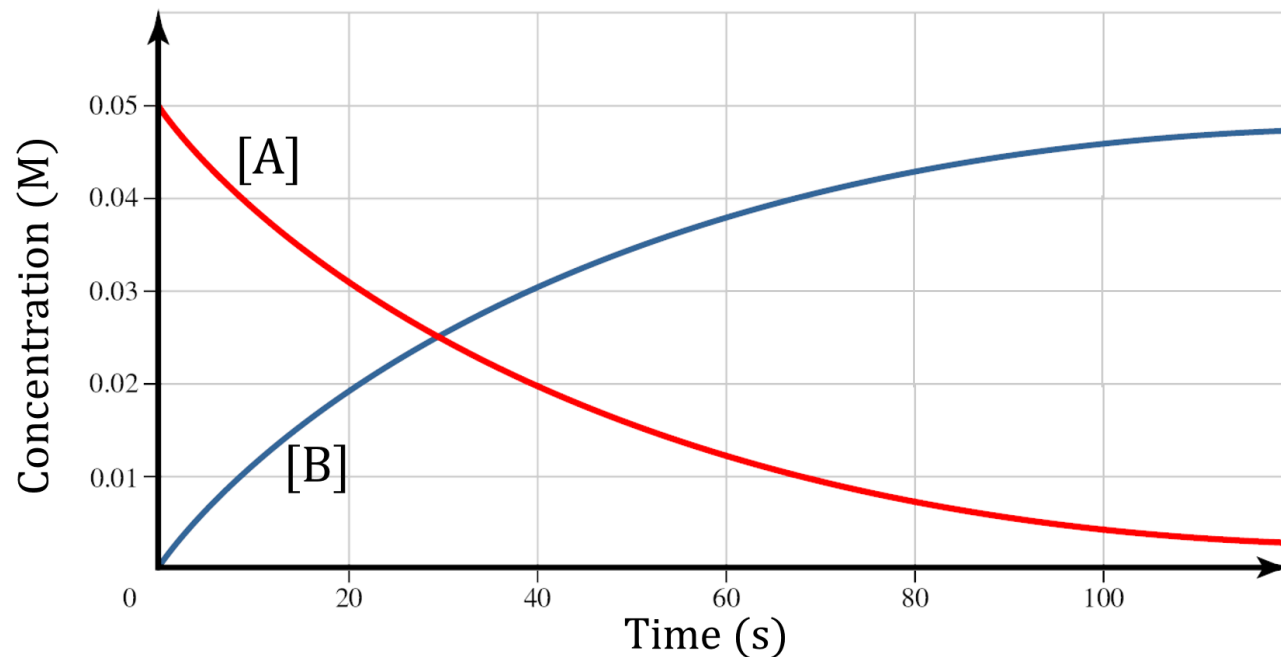


FIGURE 14.2

Rates of Reactions

14.1

We can also consider reactions that involve more than one reactant or product, such as the reaction below.



$$\text{Rate} = -\frac{1}{a} \cdot \frac{\Delta[\text{A}]}{\Delta t} = +\frac{1}{b} \cdot \frac{\Delta[\text{B}]}{\Delta t} = +\frac{1}{c} \cdot \frac{\Delta[\text{C}]}{\Delta t}$$

negative positive positive

$$\text{Rate} = -\frac{1}{1} \cdot \frac{\Delta[\text{A}]}{\Delta t} = +\frac{1}{1} \cdot \frac{\Delta[\text{B}]}{\Delta t} = +\frac{1}{2} \cdot \frac{\Delta[\text{C}]}{\Delta t}$$

Concentration (M)

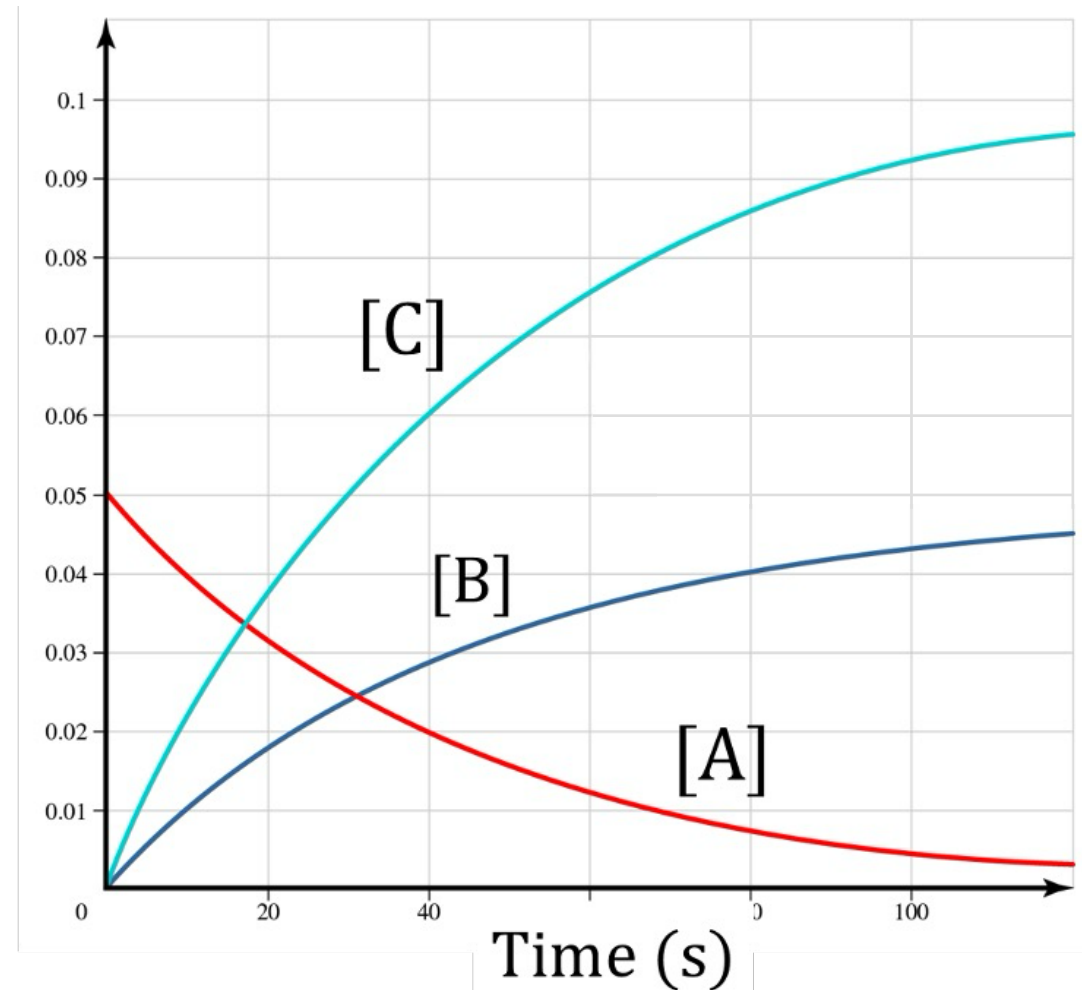


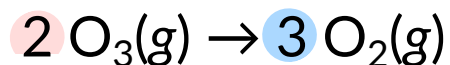
FIGURE 14.4

EXAMPLE PROBLEM

Learning Objective(s): 14.1

The conversion of ozone to oxygen was studied in an experiment and the rate of O_3 consumption was measured as $2.5 \times 10^{-5} \text{ M/s}$.

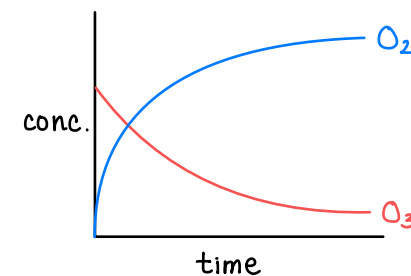
$$-\frac{\Delta[O_3]}{\Delta t} = 2.5 \times 10^{-5} \frac{\text{M}}{\text{s}}$$



$$\text{Rate} = -\underbrace{\frac{1}{2} \cdot \frac{\Delta[O_3]}{\Delta t}}_{\text{negative}} = +\underbrace{\frac{1}{3} \cdot \frac{\Delta[O_2]}{\Delta t}}_{\text{positive}}$$

What was the rate of O_2 production in this experiment?

$$\begin{aligned} -\frac{1}{2} \cdot \frac{\Delta[O_3]}{\Delta t} &= +\frac{1}{3} \cdot \frac{\Delta[O_2]}{\Delta t} \\ +\frac{\Delta[O_2]}{\Delta t} &= -\frac{3}{2} \cdot \frac{\Delta[O_3]}{\Delta t} \quad ; \quad -\frac{\Delta[O_3]}{\Delta t} = 2.5 \times 10^{-5} \frac{\text{M}}{\text{s}} \\ &= \frac{3}{2} \cdot \left(2.5 \times 10^{-5} \frac{\text{M}}{\text{s}} \right) \\ +\frac{\Delta[O_2]}{\Delta t} &= 3.75 \times 10^{-5} \frac{\text{M}}{\text{s}} \end{aligned}$$



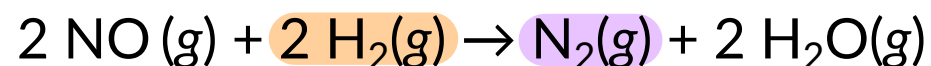
IN-CLASS QUESTION 1

$$\text{Rate} = -\frac{1}{2} \cdot \frac{\Delta[\text{NO}]}{\Delta t} = -\frac{1}{2} \cdot \frac{\Delta[\text{H}_2]}{\Delta t} = +\frac{1}{1} \cdot \frac{\Delta[\text{N}_2]}{\Delta t} = +\frac{1}{2} \cdot \frac{\Delta[\text{H}_2\text{O}]}{\Delta t}$$

Learning Objective(s): 14.1

The rate of production of $\text{N}_2(\text{g})$ in the reaction below is 0.016 M/s .

$$+\frac{\Delta[\text{N}_2]}{\Delta t}$$



$$-\frac{\Delta[\text{H}_2]}{\Delta t}$$

What is the rate of consumption of $\text{H}_2(\text{g})$ in M/s ?

$$-\frac{1}{2} \cdot \frac{\Delta[\text{H}_2]}{\Delta t} = +\frac{1}{1} \cdot \frac{\Delta[\text{N}_2]}{\Delta t} ; \quad \frac{\Delta[\text{N}_2]}{\Delta t} = 0.016 \frac{\text{M}}{\text{s}}$$

$$-\frac{\Delta[\text{H}_2]}{\Delta t} = +\frac{2}{1} \cdot \frac{\Delta[\text{N}_2]}{\Delta t}$$

$$= \frac{2}{1} \cdot \left(0.016 \frac{\text{M}}{\text{s}}\right)$$

$$-\frac{\Delta[\text{H}_2]}{\Delta t} = 0.032 \frac{\text{M}}{\text{s}}$$

Alternative solution:



-rewrite M as mol/L $0.016 \frac{\text{M}}{\text{s}} = 0.016 \frac{\text{mol N}_2}{\text{L} \cdot \text{s}}$

-use $\text{mol}:\text{mol}$ ratio $0.016 \frac{\text{mol N}_2}{\text{L} \cdot \text{s}} \times \frac{2 \text{ mol H}_2}{1 \text{ mol N}_2} = 0.032 \frac{\text{mol H}_2}{\text{L} \cdot \text{s}}$

-rewrite mol/L as M $0.032 \frac{\text{mol H}_2}{\text{L} \cdot \text{s}} = 0.032 \frac{\text{M}}{\text{s}}$

Reaction Rates and Concentration: Rate Laws

14.2

The *average rate* of a chemical reaction is the *total* change in concentration of a reactant or product over the *total* time interval.

$$\text{Rate}_{\text{avg}} = \frac{[\text{final}] - [\text{initial}]}{t_{\text{final}} - t_{\text{initial}}}$$

- Can be misleading as most reactions *slow down over time*.

The instantaneous rate of a reaction is the rate at a single moment in time.

- It is given by the *slope of a line tangent* to the curve representing the concentration over time.

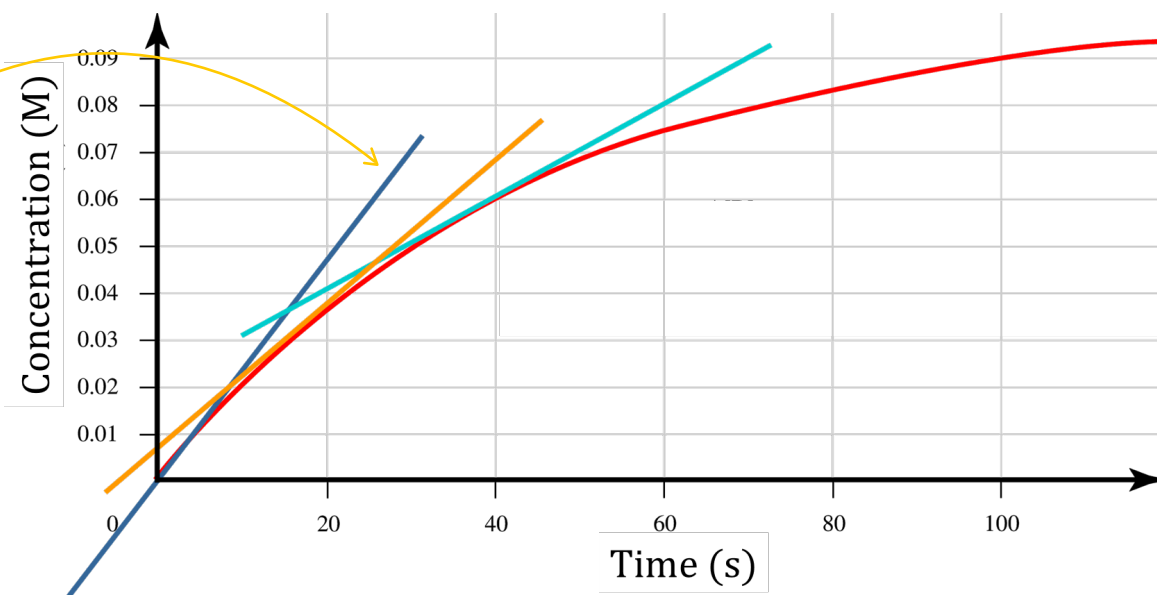
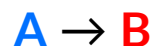


FIGURE 14.6

IN-CLASS QUESTION 2

Learning Objective(s): 14.1

For the generic reaction



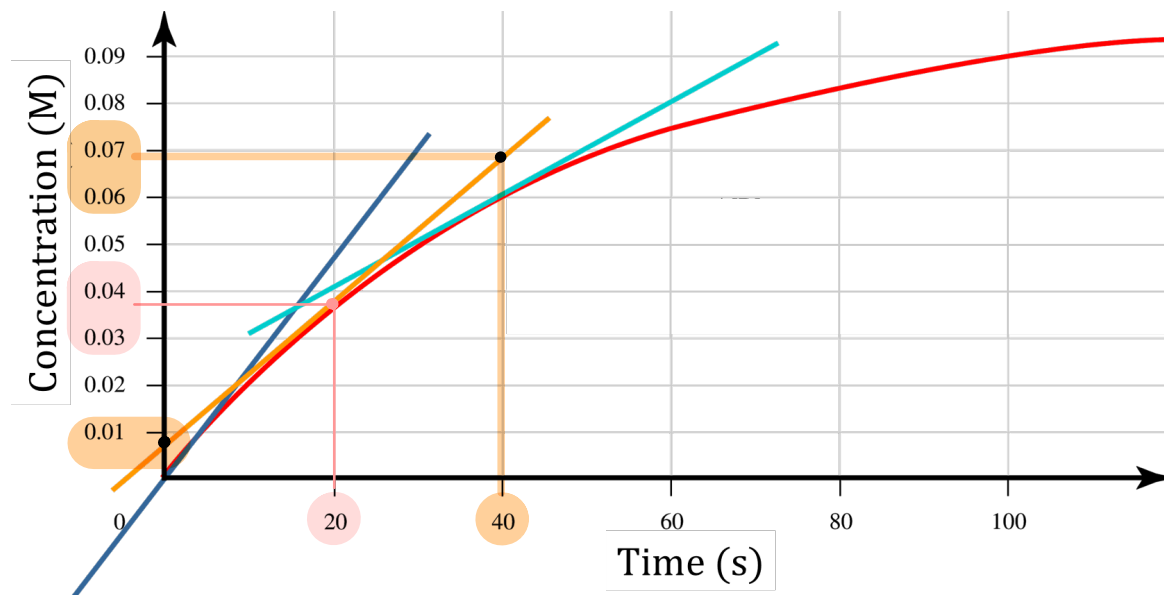
use the plot of [B] vs. time to estimate the following rates in M/s.

- a) The *average* rate of production of B from 0 to 20 seconds.

$$\text{Rate}_{\text{avg}} = \frac{[\text{B}]_{20\text{s}} - [\text{B}]_{0\text{s}}}{20\text{s} - 0\text{s}} = \frac{0.036\text{ M} - 0\text{ M}}{20\text{s} - 0\text{s}} = 0.0018 \frac{\text{M}}{\text{s}}$$

- b) The *instantaneous* rate of production of B at t = 20 seconds. *use slope of tangent*

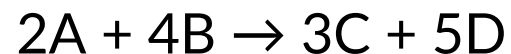
$$\text{Rate}_{\text{inst}} = \frac{\Delta[\text{B}]}{\Delta t} = \frac{0.068\text{ M} - 0.007\text{ M}}{40\text{s} - 0\text{s}} = 0.0015 \frac{\text{M}}{\text{s}}$$



BONUS QUESTION

Assessed on: Fall 2021, Exam 3

The average rate of production of C in the following chemical reaction is 8.82 mM/s at a given temperature. What is the average rate of consumption of B for this reaction in mM/s.



Reaction Rates and Concentration: Rate Laws

14.2

The initial rate is the *instantaneous* rate *as close as possible* to the *start* of the reaction (“time zero”).

- Reactions are generally *fastest* at the beginning, and slow down thereafter.
- The use of initial rates permits us to ignore the concentrations of *products* during chemical reactions.
- Chemists often measure the initial (instantaneous) rate of the reactants.

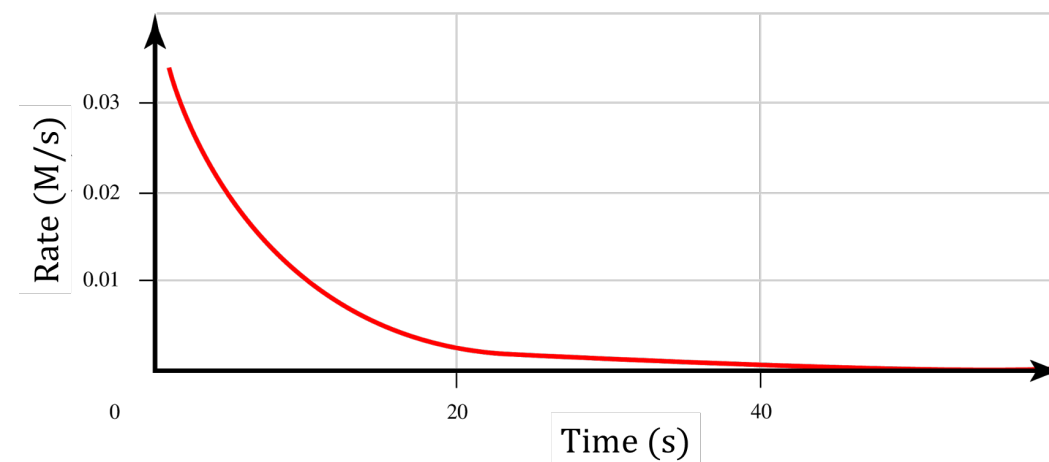


FIGURE 14.7 Instantaneous rate of a reaction over time

From this point onward, any time we use the term “*rate*” we mean the initial rate.

Reaction Rates and Concentration: Rate Laws

14.2

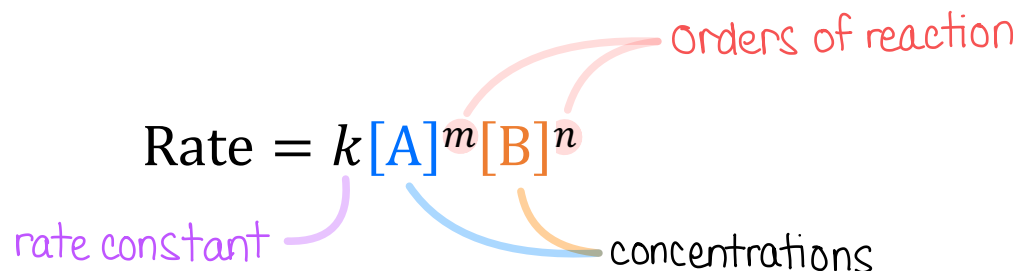
Rate law

A mathematical equation that summarizes the dependence of a reaction's rate on the concentrations of the reactants.

There are two useful forms of the rate law:

- 1) Differential rate laws (usually referred to simply as the rate law)
- 2) Integrated rate laws

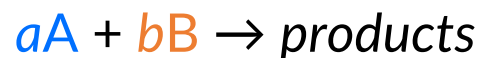
For a generic reaction: $aA + bB \rightarrow \text{products}$



Reaction Rates and Concentration: Rate Laws

14.2

For a generic reaction:



$$\text{Rate} = k[A]^m[B]^n$$

rate constant

orders

concentrations

Rate constant (k)

- A proportionality constant relating the rate to the concentrations of the reactants.
- The value of k is specific to a reaction at a particular temperature.

Orders of reaction

- Represented by powers m and n in the example above.
- The orders define how the rate is affected by the concentration of each reactant.
- Often, orders are integers or half-integers, that must be determined experimentally.